Cumulant expansion of the three-body terms

generated by `cumulant_expansion.py`

The three-body operators surviving the space restriction are decomposed into two-body and one-body operators (plus scalar terms) with density-matrix coefficients, using the spin-free cumulant decomposition of `cumulant_decomposition/spin_free/spin_free.tex`:

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx = {} & D^p_s \hat{E}^{qr}_{tu} - \tfrac{1}{2}\, D^p_t \hat{E}^{qr}_{su} - \tfrac{1}{2}\, D^p_u \hat{E}^{qr}_{ts} && (1)\\
& - \tfrac{1}{2}\, D^q_s \hat{E}^{pr}_{tu} + D^q_t \hat{E}^{pr}_{su} - \tfrac{1}{2}\, D^q_u \hat{E}^{pr}_{st} && (2)\\
& - \tfrac{1}{2}\, D^r_s \hat{E}^{qp}_{tu} - \tfrac{1}{2}\, D^r_t \hat{E}^{pq}_{su} + D^r_u \hat{E}^{pq}_{st} + D^{pq}_{st} \hat{E}^r_u && (3)\\
& - \tfrac{1}{2}\, D^{pq}_{su} \hat{E}^r_t - \tfrac{1}{2}\, D^{pq}_{ut} \hat{E}^r_s - \tfrac{1}{2}\, D^{pr}_{st} \hat{E}^q_u && (4)\\
& + D^{pr}_{su} \hat{E}^q_t - \tfrac{1}{2}\, D^{pr}_{tu} \hat{E}^q_s - \tfrac{1}{2}\, D^{qr}_{su} \hat{E}^p_t + D^{qr}_{tu} \hat{E}^p_s && (5)\\
& - \tfrac{1}{2}\, D^{rq}_{st} \hat{E}^p_u - 2\, D^p_s D^q_t \hat{E}^r_u - 2\, D^p_s D^r_u \hat{E}^q_t && (6)\\
& - \tfrac{1}{2}\, D^p_t D^q_u \hat{E}^r_s - \tfrac{1}{2}\, D^p_t D^r_s \hat{E}^q_u - \tfrac{1}{2}\, D^p_u D^q_s \hat{E}^r_t && (7)\\
& - \tfrac{1}{2}\, D^p_u D^r_t \hat{E}^q_s - \tfrac{1}{2}\, D^q_s D^r_t \hat{E}^p_u - 2\, D^q_t D^r_u \hat{E}^p_s && (8)\\
& - \tfrac{1}{2}\, D^q_u D^r_s \hat{E}^p_t + D^p_s D^q_u \hat{E}^r_t + D^p_s D^r_t \hat{E}^q_u && (9)\\
& + D^p_t D^q_s \hat{E}^r_u + D^p_t D^r_u \hat{E}^q_s + D^p_u D^q_t \hat{E}^r_s + D^p_u D^r_s \hat{E}^q_t && (10)\\
& + D^q_s D^r_u \hat{E}^p_t + D^q_t D^r_s \hat{E}^p_u + D^q_u D^r_t \hat{E}^p_s - D^p_s D^{qr}_{tu} && (11)\\
& + \tfrac{1}{2}\, D^p_t D^{qr}_{su} + \tfrac{1}{2}\, D^p_u D^{rq}_{st} + \tfrac{1}{2}\, D^q_s D^{pr}_{tu} - D^q_t D^{pr}_{su} && (12)\\
& + \tfrac{1}{2}\, D^q_u D^{pr}_{st} + \tfrac{1}{2}\, D^r_s D^{pq}_{ut} + \tfrac{1}{2}\, D^r_t D^{pq}_{su} - D^r_u D^{pq}_{st} && (13)\\
& + 4\, D^p_s D^q_t D^r_u - 2\, D^p_s D^q_u D^r_t - 2\, D^p_t D^q_s D^r_u + D^p_t D^q_u D^r_s && (14)\\
& + D^p_u D^q_s D^r_t - 2\, D^p_u D^q_t D^r_s && (15)
\end{aligned}$$

This is an operator identity: the labels p, q, r, s, t, u are fixed on both sides, so no index may be renamed; only the rows of each operator can be reordered (the parity sign goes into the coefficient). With the terms grouped by the operator in this sense, the decomposition reads

$$\begin{aligned}
\hat{E}^{pqr}_{stu} \approx {} & \hat{E}^{pq}_{st}\ (D^r_u) && (16)\\
& \hat{E}^{pq}_{su}\ \left(-\tfrac{1}{2}\, D^r_t\right) && (17)\\
& \hat{E}^{pq}_{tu}\ \left(\tfrac{1}{2}\, D^r_s\right) && (18)\\
& \hat{E}^{pr}_{st}\ \left(-\tfrac{1}{2}\, D^q_u\right) && (19)\\
& \hat{E}^{pr}_{su}\ (D^q_t) && (20)\\
& \hat{E}^{pr}_{tu}\ \left(-\tfrac{1}{2}\, D^q_s\right) && (21)\\
& \hat{E}^{qr}_{st}\ \left(\tfrac{1}{2}\, D^p_u\right) && (22)\\
& \hat{E}^{qr}_{su}\ \left(-\tfrac{1}{2}\, D^p_t\right) && (23)
\end{aligned}$$

$$\hat{E}^{qr}_{tu}\ (D^{p}_{s}) \tag{24}$$
$$\hat{E}^{p}_{s}\ (D^{qr}_{tu} - 2\,D^{q}_{t}D^{r}_{u} + D^{q}_{u}D^{r}_{t}) \tag{25}$$
$$\hat{E}^{p}_{t}\ \left(-\tfrac{1}{2}\,D^{qr}_{su} + D^{q}_{s}D^{r}_{u} - \tfrac{1}{2}\,D^{q}_{u}D^{r}_{s}\right) \tag{26}$$
$$\hat{E}^{p}_{u}\ \left(-\tfrac{1}{2}\,D^{q}_{s}D^{r}_{t} + D^{q}_{t}D^{r}_{s} - \tfrac{1}{2}\,D^{rq}_{st}\right) \tag{27}$$
$$\hat{E}^{q}_{s}\ \left(-\tfrac{1}{2}\,D^{pr}_{tu} + D^{p}_{t}D^{r}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{r}_{t}\right) \tag{28}$$
$$\hat{E}^{q}_{t}\ (D^{pr}_{su} - 2\,D^{p}_{s}D^{r}_{u} + D^{p}_{u}D^{r}_{s}) \tag{29}$$
$$\hat{E}^{q}_{u}\ \left(-\tfrac{1}{2}\,D^{pr}_{st} + D^{p}_{s}D^{r}_{t} - \tfrac{1}{2}\,D^{p}_{t}D^{r}_{s}\right) \tag{30}$$
$$\hat{E}^{r}_{s}\ \left(-\tfrac{1}{2}\,D^{pq}_{ut} - \tfrac{1}{2}\,D^{p}_{t}D^{q}_{u} + D^{p}_{u}D^{q}_{t}\right) \tag{31}$$
$$\hat{E}^{r}_{t}\ \left(-\tfrac{1}{2}\,D^{pq}_{su} + D^{p}_{s}D^{q}_{u} - \tfrac{1}{2}\,D^{p}_{u}D^{q}_{s}\right) \tag{32}$$
$$\hat{E}^{r}_{u}\ (D^{pq}_{st} - 2\,D^{p}_{s}D^{q}_{t} + D^{p}_{t}D^{q}_{s}) \tag{33}$$
$$-\,D^{p}_{s}D^{qr}_{tu} + 4\,D^{p}_{s}D^{q}_{t}D^{r}_{u} - 2\,D^{p}_{s}D^{q}_{u}D^{r}_{t} + \tfrac{1}{2}\,D^{p}_{t}D^{qr}_{su} \tag{34}$$
$$-\,2\,D^{p}_{t}D^{q}_{s}D^{r}_{u} + D^{p}_{t}D^{q}_{u}D^{r}_{s} + D^{p}_{u}D^{q}_{s}D^{r}_{t} \tag{35}$$
$$-\,2\,D^{p}_{u}D^{q}_{t}D^{r}_{s} + \tfrac{1}{2}\,D^{p}_{u}D^{rq}_{st} + \tfrac{1}{2}\,D^{q}_{s}D^{pr}_{tu} \tag{36}$$
$$-\,D^{q}_{t}D^{pr}_{su} + \tfrac{1}{2}\,D^{q}_{u}D^{pr}_{st} + \tfrac{1}{2}\,D^{r}_{s}D^{pq}_{ut} \tag{37}$$
$$+\,\tfrac{1}{2}\,D^{r}_{t}D^{pq}_{su} - D^{r}_{u}D^{pq}_{st} \tag{38}$$

(19 operator groups; a group with no $\hat{E}$ shown is the scalar part.) The $\approx$ drops the normal-ordered three-body remainder and the three-cumulant λ^{pqr}_{stu}; for a single-determinant reference $\lambda_3 = 0$ identically. Every density matrix carrying a virtual index vanishes; density matrices with occupied or whole-space indices are kept symbolic as D^{i}_{j}, D^{ij}_{kl}, ... (no delta-function substitution). The formula is verified numerically for a closed-shell single-determinant reference on Jordan–Wigner matrices (random index assignments with and without the virtual rule).

1 Final blocks

Only the blocks containing surviving three-body terms change; the three-body pieces are decomposed and merged back into the block. Blocks not shown are unchanged from `space_restriction.tex`.

$\left[\hat{V},\,\hat{T}_2\right]$

after the space restriction (2 terms, 1 three-body):

$$\left[\hat{V},\,\hat{T}_2\right] = \tfrac{1}{2}\,g^{ab}_{pq}t^{ab}_{ij}\hat{E}^{pq}_{ij} + g^{br}_{pq}t^{ab}_{ij}\hat{E}^{apq}_{ijr} \tag{39}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$g^{br}_{pq}t^{ab}_{ij}\hat{E}^{apq}_{ijr} \approx \hat{E}^{ap}_{ij}\ \left(g^{br}_{pq}t^{ab}_{ij}D^{q}_{r}\right) \tag{40}$$
$$\hat{E}^{ap}_{ir}\ \left(-\tfrac{1}{2}\,g^{br}_{pq}t^{ab}_{ij}D^{q}_{j}\right) \tag{41}$$

$$\hat{E}^{ap}_{jr}\ \left(\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{q}_{i}\right) \tag{42}$$

$$\hat{E}^{aq}_{ij}\ \left(-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{p}_{r}\right) \tag{43}$$

$$\hat{E}^{aq}_{ir}\ \left(g^{br}_{pq} t^{ab}_{ij} D^{p}_{j}\right) \tag{44}$$

$$\hat{E}^{aq}_{jr}\ \left(-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{p}_{i}\right) \tag{45}$$

$$\hat{E}^{a}_{i}\ \Big(g^{br}_{pq} t^{ab}_{ij} D^{pq}_{jr} - 2\, g^{br}_{pq} t^{ab}_{ij} D^{p}_{j} D^{q}_{r} \tag{46}$$

$$+ g^{br}_{pq} t^{ab}_{ij} D^{p}_{r} D^{q}_{j}\Big) \tag{47}$$

$$\hat{E}^{a}_{j}\ \Big(-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{pq}_{ir} + g^{br}_{pq} t^{ab}_{ij} D^{p}_{i} D^{q}_{r} \tag{48}$$

$$-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{p}_{r} D^{q}_{i}\Big) \tag{49}$$

$$\hat{E}^{a}_{r}\ \Big(-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{p}_{i} D^{q}_{j} + g^{br}_{pq} t^{ab}_{ij} D^{p}_{j} D^{q}_{i} \tag{50}$$

$$-\tfrac{1}{2}\, g^{br}_{pq} t^{ab}_{ij} D^{qp}_{ij}\Big) \tag{51}$$

the decomposition of the three-body part produces 15 terms; the final block (surviving terms and decomposition merged) has 16 terms:

$$\Big[\hat{V},\, \hat{T}_2\Big] = g^{bp}_{qr} t^{ab}_{ij} D^{jp}_{qr} \hat{E}^{a}_{i} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{jp}_{qr} \hat{E}^{a}_{i} \tag{52}$$

$$- 2\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{p} D^{q}_{r} \hat{E}^{a}_{i} + g^{bq}_{pr} t^{ab}_{ji} D^{j}_{p} D^{q}_{r} \hat{E}^{a}_{i} \tag{53}$$

$$+ g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r} D^{j}_{p} \hat{E}^{a}_{i} - \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{q}_{r} D^{j}_{p} \hat{E}^{a}_{i} \tag{54}$$

$$- \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ij} D^{i}_{q} D^{j}_{r} \hat{E}^{a}_{p} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{ij}_{qr} \hat{E}^{a}_{p} \tag{55}$$

$$+ g^{bp}_{qr} t^{ab}_{ji} D^{i}_{q} D^{j}_{r} \hat{E}^{a}_{p} + g^{bq}_{pr} t^{ab}_{ij} D^{q}_{r} \hat{E}^{ap}_{ij} \tag{56}$$

$$- \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r} \hat{E}^{ap}_{ij} - \tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{r} \hat{E}^{ap}_{iq} \tag{57}$$

$$+ \tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ji} D^{j}_{r} \hat{E}^{ap}_{iq} + g^{bq}_{rp} t^{ab}_{ij} D^{j}_{r} \hat{E}^{ap}_{iq} \tag{58}$$

$$- \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{j}_{r} \hat{E}^{ap}_{iq} + \tfrac{1}{2}\, g^{ab}_{pq} t^{ab}_{ij} \hat{E}^{pq}_{ij} \tag{59}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (5 operator groups):

$$\Big[\hat{V},\, \hat{T}_2\Big] = \hat{E}^{a}_{i}\ \Big(g^{bp}_{qr} t^{ab}_{ij} D^{jp}_{qr} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{jp}_{qr} \tag{60}$$

$$-2\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{p} D^{q}_{r} + g^{bq}_{pr} t^{ab}_{ji} D^{j}_{p} D^{q}_{r} \tag{61}$$

$$+ g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r} D^{j}_{p} - \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{q}_{r} D^{j}_{p}\Big) \tag{62}$$

$$\hat{E}^{a}_{p}\ \Big(-\tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ij} D^{i}_{q} D^{j}_{r} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{ij}_{qr} \tag{63}$$

$$+ g^{bp}_{qr} t^{ab}_{ji} D^{i}_{q} D^{j}_{r}\Big) \tag{64}$$

$$\hat{E}^{pa}_{ij}\ \Big(-g^{bq}_{pr} t^{ab}_{ij} D^{q}_{r} + \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r}\Big) \tag{65}$$

$$\hat{E}^{pa}_{qi}\ \Big(-\tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{r} + \tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ji} D^{j}_{r} \tag{66}$$

$$+ g^{bq}_{rp} t^{ab}_{ij} D^{j}_{r} - \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{j}_{r}\Big) \tag{67}$$

$$\hat{E}^{pq}_{ij}\ \left(\tfrac{1}{2}\, g^{ab}_{pq} t^{ab}_{ij}\right) \tag{68}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^{\dagger},\, \hat{T}_2\right]$$

after the space restriction (6 terms, 4 three-body):

$$\left[\left[\hat{F},\, \hat{T}_2\right]^{\dagger},\, \hat{T}_2\right] = f^{a}_{b} t^{ac}_{ij} t^{bc}_{kl} \hat{E}^{ij}_{kl} \tag{69}$$

$$- f^{l}_{p} t^{ab}_{il} t^{ab}_{jk} \hat{E}^{ip}_{jk} - f^{c}_{d} t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{bkl} \tag{70}$$

$$- f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{klp} + f^{l}_{p} t^{bc}_{il} t^{ac}_{jk} \hat{E}^{aip}_{bjk} \tag{71}$$

$$- f^{l}_{p} t^{bc}_{li} t^{ac}_{jk} \hat{E}^{aip}_{bjk} \tag{72}$$

each three-body term is decomposed with the grouped identity. Every template index appears in exactly one factor of each piece, so a whole operator group survives or dies together: a group survives iff its operator carries every virtual index of the pattern; in particular the scalar part (pure density matrices) always vanishes. The surviving groups are: 9 of the 19 groups survive (6 two-body, 3 one-body):

$$-f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{klp} \approx \hat{E}^{ai}_{kl}\ \left(-f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{j}_{p}\right) \tag{73}$$

$$\hat{E}^{ai}_{kp}\ \left(\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{j}_{l}\right) \tag{74}$$

$$\hat{E}^{ai}_{lp}\ \left(-\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{j}_{k}\right) \tag{75}$$

$$\hat{E}^{aj}_{kl}\ \left(\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{p}\right) \tag{76}$$

$$\hat{E}^{aj}_{kp}\ \left(-f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{l}\right) \tag{77}$$

$$\hat{E}^{aj}_{lp}\ \left(\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{k}\right) \tag{78}$$

$$\hat{E}^{a}_{k}\ \left(-f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{ij}_{lp} + 2\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{l} D^{j}_{p}\right. \tag{79}$$

$$\left. - f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{p} D^{j}_{l}\right) \tag{80}$$

$$\hat{E}^{a}_{l}\ \left(\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{ij}_{kp}\right. \tag{81}$$

$$\left. - f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{k} D^{j}_{p} + \tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{p} D^{j}_{k}\right) \tag{82}$$

$$\hat{E}^{a}_{p}\ \left(\tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{k} D^{j}_{l}\right. \tag{83}$$

$$\left. - f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{i}_{l} D^{j}_{k} + \tfrac{1}{2}\, f^{b}_{p} t^{bc}_{ij} t^{ac}_{kl} D^{ji}_{kl}\right) \tag{84}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-f^{c}_{d} t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{bkl} \approx \hat{E}^{ai}_{bk}\ \left(-f^{c}_{d} t^{bd}_{ij} t^{ac}_{kl} D^{j}_{l}\right) \tag{85}$$

$$\hat{E}^{ai}_{bl}\ \left(\tfrac{1}{2}\, f^{c}_{d} t^{bd}_{ij} t^{ac}_{kl} D^{j}_{k}\right) \tag{86}$$

$$\hat{E}^{aj}_{bk}\ \left(\tfrac{1}{2}\, f^{c}_{d} t^{bd}_{ij} t^{ac}_{kl} D^{i}_{l}\right) \tag{87}$$

$$\hat{E}^{aj}_{bl}\ \left(-f^{c}_{d}t^{bd}_{ij}t^{ac}_{kl}D^{i}_{k}\right) \tag{88}$$

$$\hat{E}^{a}_{b}\ \left(-f^{c}_{d}t^{bd}_{ij}t^{ac}_{kl}D^{ij}_{kl}+2\,f^{c}_{d}t^{bd}_{ij}t^{ac}_{kl}D^{i}_{k}D^{j}_{l}\right. \tag{89}$$

$$\left.-f^{c}_{d}t^{bd}_{ij}t^{ac}_{kl}D^{i}_{l}D^{j}_{k}\right) \tag{90}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$-f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}\hat{E}^{aip}_{bjk}\approx\hat{E}^{ai}_{bj}\ \left(-f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{p}_{k}\right) \tag{91}$$

$$\hat{E}^{ai}_{bk}\ \left(\tfrac{1}{2}\,f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{p}_{j}\right) \tag{92}$$

$$\hat{E}^{ap}_{bj}\ \left(\tfrac{1}{2}\,f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{i}_{k}\right) \tag{93}$$

$$\hat{E}^{ap}_{bk}\ \left(-f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{i}_{j}\right) \tag{94}$$

$$\hat{E}^{a}_{b}\ \left(-f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{ip}_{jk}+2\,f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{i}_{j}D^{p}_{k}\right. \tag{95}$$

$$\left.-f^{l}_{p}t^{bc}_{li}t^{ac}_{jk}D^{i}_{k}D^{p}_{j}\right) \tag{96}$$

5 of the 19 groups survive (4 two-body, 1 one-body):

$$f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}\hat{E}^{aip}_{bjk}\approx\hat{E}^{ai}_{bj}\ \left(f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{p}_{k}\right) \tag{97}$$

$$\hat{E}^{ai}_{bk}\ \left(-\tfrac{1}{2}\,f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{p}_{j}\right) \tag{98}$$

$$\hat{E}^{ap}_{bj}\ \left(-\tfrac{1}{2}\,f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{i}_{k}\right) \tag{99}$$

$$\hat{E}^{ap}_{bk}\ \left(f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{i}_{j}\right) \tag{100}$$

$$\hat{E}^{a}_{b}\ \left(f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{ip}_{jk}-2\,f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{i}_{j}D^{p}_{k}\right. \tag{101}$$

$$\left.+f^{l}_{p}t^{bc}_{il}t^{ac}_{jk}D^{i}_{k}D^{p}_{j}\right) \tag{102}$$

the decomposition of the three-body part produces 36 terms; the final block (surviving terms and decomposition merged) has 38 terms:

$$\left[\left[\hat{F},\,\hat{T}_{2}\right]^{\dagger},\,\hat{T}_{2}\right]=2\,f^{c}_{d}t^{bd}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{l}\hat{E}^{a}_{b} \tag{103}$$

$$-f^{c}_{d}t^{bd}_{kl}t^{ac}_{ij}D^{ij}_{kl}\hat{E}^{a}_{b}-f^{c}_{d}t^{bd}_{lj}t^{ac}_{ik}D^{k}_{l}D^{i}_{j}\hat{E}^{a}_{b} \tag{104}$$

$$-2\,f^{l}_{p}t^{bc}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{p}\hat{E}^{a}_{b}+f^{l}_{p}t^{bc}_{jl}t^{ac}_{ki}D^{i}_{j}D^{k}_{p}\hat{E}^{a}_{b} \tag{105}$$

$$+f^{l}_{p}t^{bc}_{kl}t^{ac}_{ij}D^{ij}_{kp}\hat{E}^{a}_{b}+2\,f^{l}_{p}t^{bc}_{lj}t^{ac}_{ik}D^{i}_{j}D^{k}_{p}\hat{E}^{a}_{b} \tag{106}$$

$$-f^{l}_{p}t^{bc}_{lj}t^{ac}_{ki}D^{i}_{j}D^{k}_{p}\hat{E}^{a}_{b}-f^{l}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{ij}_{kp}\hat{E}^{a}_{b} \tag{107}$$

$$-f^{b}_{p}t^{bc}_{jk}t^{ac}_{il}D^{jk}_{lp}\hat{E}^{a}_{i}+\tfrac{1}{2}\,f^{b}_{p}t^{bc}_{jk}t^{ac}_{li}D^{jk}_{lp}\hat{E}^{a}_{i} \tag{108}$$

$$+2\,f^{b}_{p}t^{bc}_{kl}t^{ac}_{ij}D^{j}_{k}D^{l}_{p}\hat{E}^{a}_{i}-f^{b}_{p}t^{bc}_{kl}t^{ac}_{ji}D^{j}_{k}D^{l}_{p}\hat{E}^{a}_{i} \tag{109}$$

$$-f^{b}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{l}_{p}D^{j}_{k}\hat{E}^{a}_{i}+\tfrac{1}{2}\,f^{b}_{p}t^{bc}_{lk}t^{ac}_{ji}D^{l}_{p}D^{j}_{k}\hat{E}^{a}_{i} \tag{110}$$

$$+\tfrac{1}{2}\,f^{b}_{p}t^{bc}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{l}\hat{E}^{a}_{p}-f^{b}_{p}t^{bc}_{lj}t^{ac}_{ik}D^{k}_{l}D^{i}_{j}\hat{E}^{a}_{p} \tag{111}$$

$$+\tfrac{1}{2}\,f^{b}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{ij}_{kl}\hat{E}^{a}_{p}-f^{c}_{d}t^{bd}_{il}t^{ac}_{jk}D^{k}_{l}\hat{E}^{ai}_{bj} \tag{112}$$

$$+\tfrac{1}{2}\,f_d^c t_{il}^{bd} t_{kj}^{ac} D_l^k \hat{E}_{bj}^{ai} + \tfrac{1}{2}\,f_d^c t_{li}^{bd} t_{jk}^{ac} D_l^k \hat{E}_{bj}^{ai} \tag{113}$$

$$-\,f_d^c t_{li}^{bd} t_{kj}^{ac} D_l^k \hat{E}_{bj}^{ai} + f_p^l t_{il}^{bc} t_{jk}^{ac} D_p^k \hat{E}_{bj}^{ai} \tag{114}$$

$$-\tfrac{1}{2}\,f_p^l t_{il}^{bc} t_{kj}^{ac} D_p^k \hat{E}_{bj}^{ai} - f_p^l t_{li}^{bc} t_{jk}^{ac} D_p^k \hat{E}_{bj}^{ai} \tag{115}$$

$$+\tfrac{1}{2}\,f_p^l t_{li}^{bc} t_{kj}^{ac} D_p^k \hat{E}_{bj}^{ai} - f_p^b t_{il}^{bc} t_{jk}^{ac} D_p^l \hat{E}_{jk}^{ai} \tag{116}$$

$$+\tfrac{1}{2}\,f_p^b t_{li}^{bc} t_{jk}^{ac} D_p^l \hat{E}_{jk}^{ai} + \tfrac{1}{2}\,f_p^b t_{il}^{bc} t_{jk}^{ac} D_l^k \hat{E}_{jp}^{ai} \tag{117}$$

$$-\tfrac{1}{2}\,f_p^b t_{il}^{bc} t_{kj}^{ac} D_l^k \hat{E}_{jp}^{ai} - f_p^b t_{li}^{bc} t_{jk}^{ac} D_l^k \hat{E}_{jp}^{ai} \tag{118}$$

$$+\tfrac{1}{2}\,f_p^b t_{li}^{bc} t_{kj}^{ac} D_l^k \hat{E}_{jp}^{ai} - \tfrac{1}{2}\,f_p^l t_{kl}^{bc} t_{ij}^{ac} D_k^j \hat{E}_{bi}^{ap} \tag{119}$$

$$+\,f_p^l t_{kl}^{bc} t_{ji}^{ac} D_k^j \hat{E}_{bi}^{ap} + \tfrac{1}{2}\,f_p^l t_{lk}^{bc} t_{ij}^{ac} D_k^j \hat{E}_{bi}^{ap} \tag{120}$$

$$-\,f_p^l t_{lk}^{bc} t_{ji}^{ac} D_k^j \hat{E}_{bi}^{ap} + f_b^a t_{ij}^{ac} t_{kl}^{bc} \hat{E}_{kl}^{ij} \tag{121}$$

$$-\,f_p^l t_{il}^{ab} t_{jk}^{ab} \hat{E}_{jk}^{ip} \tag{122}$$

grouped by the operator, so that all coefficient pieces multiplying the same $\hat{E}$ can be substituted together (9 operator groups):

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = \hat{E}_b^a\,\Big(2\,f_d^c t_{jl}^{bd} t_{ik}^{ac} D_j^i D_l^k - f_d^c t_{kl}^{bd} t_{ij}^{ac} D_{kl}^{ij} \tag{123}$$

$$-f_d^c t_{lj}^{bd} t_{ik}^{ac} D_l^k D_j^i - 2\,f_p^l t_{jl}^{bc} t_{ik}^{ac} D_j^i D_p^k \tag{124}$$

$$+f_p^l t_{jl}^{bc} t_{ki}^{ac} D_j^i D_p^k + f_p^l t_{kl}^{bc} t_{ij}^{ac} D_{kp}^{ij} \tag{125}$$

$$+2\,f_p^l t_{lj}^{bc} t_{ik}^{ac} D_j^i D_p^k - f_p^l t_{lj}^{bc} t_{ki}^{ac} D_j^i D_p^k \tag{126}$$

$$-f_p^l t_{lk}^{bc} t_{ij}^{ac} D_{kp}^{ij}\Big) \tag{127}$$

$$\hat{E}_i^a\,\Big(-f_p^b t_{jk}^{bc} t_{il}^{ac} D_{lp}^{jk} + \tfrac{1}{2}\,f_p^b t_{jk}^{bc} t_{li}^{ac} D_{lp}^{jk} \tag{128}$$

$$+2\,f_p^b t_{kl}^{bc} t_{ij}^{ac} D_k^j D_p^l - f_p^b t_{kl}^{bc} t_{ji}^{ac} D_k^j D_p^l \tag{129}$$

$$-f_p^b t_{lk}^{bc} t_{ij}^{ac} D_p^l D_k^j + \tfrac{1}{2}\,f_p^b t_{lk}^{bc} t_{ji}^{ac} D_p^l D_k^j\Big) \tag{130}$$

$$\hat{E}_p^a\,\Big(\tfrac{1}{2}\,f_p^b t_{jl}^{bc} t_{ik}^{ac} D_j^i D_l^k \tag{131}$$

$$-f_p^b t_{lj}^{bc} t_{ik}^{ac} D_l^k D_j^i + \tfrac{1}{2}\,f_p^b t_{lk}^{bc} t_{ij}^{ac} D_{kl}^{ij}\Big) \tag{132}$$

$$\hat{E}_{jb}^{ia}\,\Big(-f_d^c t_{il}^{bd} t_{jk}^{ac} D_l^k + \tfrac{1}{2}\,f_d^c t_{il}^{bd} t_{kj}^{ac} D_l^k \tag{133}$$

$$+\tfrac{1}{2}\,f_d^c t_{li}^{bd} t_{jk}^{ac} D_l^k - f_d^c t_{li}^{bd} t_{kj}^{ac} D_l^k \tag{134}$$

$$+f_p^l t_{il}^{bc} t_{jk}^{ac} D_p^k - \tfrac{1}{2}\,f_p^l t_{il}^{bc} t_{kj}^{ac} D_p^k \tag{135}$$

$$-f_p^l t_{li}^{bc} t_{jk}^{ac} D_p^k + \tfrac{1}{2}\,f_p^l t_{li}^{bc} t_{kj}^{ac} D_p^k\Big) \tag{136}$$

$$\hat{E}_{jk}^{ia}\,\Big(f_p^b t_{il}^{bc} t_{jk}^{ac} D_p^l - \tfrac{1}{2}\,f_p^b t_{li}^{bc} t_{jk}^{ac} D_p^l\Big) \tag{137}$$

$$\hat{E}_{pj}^{ia}\,\Big(\tfrac{1}{2}\,f_p^b t_{il}^{bc} t_{jk}^{ac} D_l^k \tag{138}$$

$$-\tfrac{1}{2}\,f_p^b t_{il}^{bc} t_{kj}^{ac} D_l^k - f_p^b t_{li}^{bc} t_{jk}^{ac} D_l^k \tag{139}$$

$$+\tfrac{1}{2}\,f_p^b t_{li}^{bc} t_{kj}^{ac} D_l^k\Big) \tag{140}$$

$$\hat{E}_{kl}^{ij}\,\Big(f_b^a t_{ij}^{ac} t_{kl}^{bc}\Big) \tag{141}$$

$$\hat{E}^{pa}_{ib} \left(-\tfrac{1}{2}\, f^{l}_{p} t^{bc}_{kl} t^{ac}_{ij} D^{j}_{k} + f^{l}_{p} t^{bc}_{kl} t^{ac}_{ji} D^{j}_{k} \right. \tag{142}$$

$$\left. + \tfrac{1}{2}\, f^{l}_{p} t^{bc}_{lk} t^{ac}_{ij} D^{j}_{k} - f^{l}_{p} t^{bc}_{lk} t^{ac}_{ji} D^{j}_{k} \right) \tag{143}$$

$$\hat{E}^{pi}_{jk} \left(f^{l}_{p} t^{ab}_{il} t^{ab}_{jk} \right) \tag{144}$$

2 All final blocks

$\left[\hat{F},\, \hat{T}_1\right]$ **(2 terms)**

$$\left[\hat{F},\, \hat{T}_1\right] = -f^{i}_{p} t^{a}_{i} \hat{E}^{a}_{p} + f^{a}_{p} t^{a}_{i} \hat{E}^{p}_{i} \tag{145}$$

$\left[\hat{F},\, \hat{T}_2\right]$ **(1 terms)**

$$\left[\hat{F},\, \hat{T}_2\right] = f^{b}_{p} t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{146}$$

$\left[\hat{V},\, \hat{T}_1\right]$ **(2 terms)**

$$\left[\hat{V},\, \hat{T}_1\right] = -g^{ip}_{qr} t^{a}_{i} \hat{E}^{ap}_{qr} + g^{ar}_{pq} t^{a}_{i} \hat{E}^{pq}_{ir} \tag{147}$$

$\left[\hat{V},\, \hat{T}_2\right]$ **(16 terms)**

$$\left[\hat{V},\, \hat{T}_2\right] = g^{bp}_{qr} t^{ab}_{ij} D^{jp}_{qr} \hat{E}^{a}_{i} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{jp}_{qr} \hat{E}^{a}_{i} \tag{148}$$

$$- 2\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{p} D^{q}_{r} \hat{E}^{a}_{i} + g^{bq}_{pr} t^{ab}_{ji} D^{j}_{p} D^{q}_{r} \hat{E}^{a}_{i} \tag{149}$$

$$+ g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r} D^{j}_{p} \hat{E}^{a}_{i} - \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{q}_{r} D^{j}_{p} \hat{E}^{a}_{i} \tag{150}$$

$$- \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ij} D^{i}_{q} D^{j}_{r} \hat{E}^{a}_{p} - \tfrac{1}{2}\, g^{bp}_{qr} t^{ab}_{ji} D^{ij}_{qr} \hat{E}^{a}_{p} \tag{151}$$

$$+ g^{bp}_{qr} t^{ab}_{ji} D^{i}_{q} D^{j}_{r} \hat{E}^{a}_{p} + g^{bq}_{pr} t^{ab}_{ij} D^{q}_{r} \hat{E}^{ap}_{ij} \tag{152}$$

$$- \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ij} D^{q}_{r} \hat{E}^{ap}_{ij} - \tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ij} D^{j}_{r} \hat{E}^{ap}_{iq} \tag{153}$$

$$+ \tfrac{1}{2}\, g^{bq}_{pr} t^{ab}_{ji} D^{j}_{r} \hat{E}^{ap}_{iq} + g^{bq}_{rp} t^{ab}_{ij} D^{j}_{r} \hat{E}^{ap}_{iq} \tag{154}$$

$$- \tfrac{1}{2}\, g^{bq}_{rp} t^{ab}_{ji} D^{j}_{r} \hat{E}^{ap}_{iq} + \tfrac{1}{2}\, g^{ab}_{pq} t^{ab}_{ij} \hat{E}^{pq}_{ij} \tag{155}$$

$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_1\right]$ **(1 terms)**

$$\left[\left[\hat{F},\, \hat{T}_1\right],\, \hat{T}_1\right] = -2\, f^{b}_{j} t^{b}_{i} t^{a}_{j} \hat{E}^{a}_{i} \tag{156}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right]$ **(4 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^i_j t^b_j t^a_i \hat{E}^a_b - f^b_p t^b_i t^a_i \hat{E}^a_p \tag{157}$$

$$+ f^a_b t^a_i t^b_j \hat{E}^i_j - f^j_p t^a_j t^a_i \hat{E}^p_i \tag{158}$$

$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right]$ **(1 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] = f^c_k t^a_k t^{bc}_{ij} \hat{E}^{ab}_{ij} \tag{159}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right]$ **(2 terms)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f^b_c t^c_i t^{ab}_{jk} \hat{E}^{ai}_{jk} \tag{160}$$

$$- f^k_p t^b_k t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{161}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right]$ **(1 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = f^c_k t^{bc}_{ij} t^a_k \hat{E}^{ab}_{ij} \tag{162}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right]$ **(6 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = f^c_p t^{bc}_{ij} t^a_j \hat{E}^{ai}_{bp} \tag{163}$$

$$- f^c_p t^{bc}_{ji} t^a_j \hat{E}^{ai}_{bp} + f^b_c t^{ab}_{ij} t^c_k \hat{E}^{ij}_{ak} - f^a_p t^{ab}_{ij} t^b_k \hat{E}^{ij}_{kp} \tag{164}$$

$$- f^k_p t^{ab}_{ik} t^b_j \hat{E}^{ip}_{aj} + f^k_p t^{ab}_{ki} t^b_j \hat{E}^{ip}_{aj} \tag{165}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right]$ **(0 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 0 \tag{166}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right]$ **(38 terms)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = 2\, f^c_d t^{bd}_{jl} t^{ac}_{ik} D^i_j D^k_l \hat{E}^a_b \tag{167}$$

$$- f_d^c t_{kl}^{bd} t_{ij}^{ac} D_{kl}^{ij} \hat{E}_b^a - f_d^c t_{lj}^{bd} t_{ik}^{ac} D_l^k D_j^i \hat{E}_b^a \tag{168}$$
$$- 2\, f_p^l t_{jl}^{bc} t_{ik}^{ac} D_j^i D_p^k \hat{E}_b^a + f_p^l t_{jl}^{bc} t_{ki}^{ac} D_j^i D_p^k \hat{E}_b^a \tag{169}$$
$$+ f_p^l t_{kl}^{bc} t_{ij}^{ac} D_{kp}^{ij} \hat{E}_b^a + 2\, f_p^l t_{lj}^{bc} t_{ik}^{ac} D_j^i D_p^k \hat{E}_b^a \tag{170}$$
$$- f_p^l t_{lj}^{bc} t_{ki}^{ac} D_j^i D_p^k \hat{E}_b^a - f_p^l t_{lk}^{bc} t_{ij}^{ac} D_{kp}^{ij} \hat{E}_b^a \tag{171}$$
$$- f_p^b t_{jk}^{bc} t_{il}^{ac} D_{lp}^{jk} \hat{E}_i^a + \tfrac{1}{2}\, f_p^b t_{jk}^{bc} t_{li}^{ac} D_{lp}^{jk} \hat{E}_i^a \tag{172}$$
$$+ 2\, f_p^b t_{kl}^{bc} t_{ij}^{ac} D_k^j D_p^l \hat{E}_i^a - f_p^b t_{kl}^{bc} t_{ji}^{ac} D_k^j D_p^l \hat{E}_i^a \tag{173}$$
$$- f_p^b t_{lk}^{bc} t_{ij}^{ac} D_p^l D_k^j \hat{E}_i^a + \tfrac{1}{2}\, f_p^b t_{lk}^{bc} t_{ji}^{ac} D_p^l D_k^j \hat{E}_i^a \tag{174}$$
$$+ \tfrac{1}{2}\, f_p^b t_{jl}^{bc} t_{ik}^{ac} D_j^i D_l^k \hat{E}_p^a - f_p^b t_{lj}^{bc} t_{ik}^{ac} D_l^k D_j^i \hat{E}_p^a \tag{175}$$
$$+ \tfrac{1}{2}\, f_p^b t_{lk}^{bc} t_{ij}^{ac} D_{kl}^{ij} \hat{E}_p^a - f_d^c t_{il}^{bd} t_{jk}^{ac} D_l^k \hat{E}_{bj}^{ai} \tag{176}$$
$$+ \tfrac{1}{2}\, f_d^c t_{il}^{bd} t_{kj}^{ac} D_l^k \hat{E}_{bj}^{ai} + \tfrac{1}{2}\, f_d^c t_{li}^{bd} t_{jk}^{ac} D_l^k \hat{E}_{bj}^{ai} \tag{177}$$
$$- f_d^c t_{li}^{bd} t_{kj}^{ac} D_l^k \hat{E}_{bj}^{ai} + f_p^l t_{il}^{bc} t_{jk}^{ac} D_p^k \hat{E}_{bj}^{ai} \tag{178}$$
$$- \tfrac{1}{2}\, f_p^l t_{il}^{bc} t_{kj}^{ac} D_p^k \hat{E}_{bj}^{ai} - f_p^l t_{li}^{bc} t_{jk}^{ac} D_p^k \hat{E}_{bj}^{ai} \tag{179}$$
$$+ \tfrac{1}{2}\, f_p^l t_{li}^{bc} t_{kj}^{ac} D_p^k \hat{E}_{bj}^{ai} - f_p^b t_{il}^{bc} t_{jk}^{ac} D_p^l \hat{E}_{jk}^{ai} \tag{180}$$
$$+ \tfrac{1}{2}\, f_p^b t_{li}^{bc} t_{jk}^{ac} D_p^l \hat{E}_{jk}^{ai} + \tfrac{1}{2}\, f_p^b t_{il}^{bc} t_{jk}^{ac} D_l^k \hat{E}_{jp}^{ai} \tag{181}$$
$$- \tfrac{1}{2}\, f_p^b t_{il}^{bc} t_{kj}^{ac} D_l^k \hat{E}_{jp}^{ai} - f_p^b t_{li}^{bc} t_{jk}^{ac} D_l^k \hat{E}_{jp}^{ai} \tag{182}$$
$$+ \tfrac{1}{2}\, f_p^b t_{li}^{bc} t_{kj}^{ac} D_l^k \hat{E}_{jp}^{ai} - \tfrac{1}{2}\, f_p^l t_{kl}^{bc} t_{ij}^{ac} D_k^j \hat{E}_{bi}^{ap} \tag{183}$$
$$+ f_p^l t_{kl}^{bc} t_{ji}^{ac} D_k^j \hat{E}_{bi}^{ap} + \tfrac{1}{2}\, f_p^l t_{lk}^{bc} t_{ij}^{ac} D_k^j \hat{E}_{bi}^{ap} \tag{184}$$
$$- f_p^l t_{lk}^{bc} t_{ji}^{ac} D_k^j \hat{E}_{bi}^{ap} + f_b^a t_{ij}^{ac} t_{kl}^{bc} \hat{E}_{kl}^{ij} \tag{185}$$
$$- f_p^l t_{il}^{ab} t_{jk}^{ab} \hat{E}_{jk}^{ip} \tag{186}$$

3 All final blocks, grouped by operator

Every block with the coefficient pieces collected per operator, ready for substitution.

$\left[\hat{F},\, \hat{T}_1\right]$ **(2 terms, 2 operator groups)**

$$\left[\hat{F},\, \hat{T}_1\right] = \hat{E}_p^a \, \left(-f_p^i t_i^a\right) \tag{187}$$
$$\hat{E}_i^p \, \left(f_p^a t_i^a\right) \tag{188}$$

$\left[\hat{F},\, \hat{T}_2\right]$ **(1 terms, 1 operator groups)**

$$\left[\hat{F},\, \hat{T}_2\right] = \hat{E}_{ij}^{pa} \, \left(-f_p^b t_{ij}^{ab}\right) \tag{189}$$

$\left[\hat{V},\, \hat{T}_1\right]$ **(2 terms, 2 operator groups)**

$$\left[\hat{V},\, \hat{T}_1\right] = \hat{E}_{qr}^{pa} \, \left(g_{qr}^{ip} t_i^a\right) \tag{190}$$

$$\hat{E}^{pq}_{ri}\ \left(-g^{ar}_{pq}t^{a}_{i}\right) \tag{191}$$

$\left[\hat{V},\,\hat{T}_2\right]$ **(16 terms, 5 operator groups)**

$$\left[\hat{V},\,\hat{T}_2\right] = \hat{E}^{a}_{i}\ \Big(g^{bp}_{qr}t^{ab}_{ij}D^{jp}_{qr} - \tfrac{1}{2}\,g^{bp}_{qr}t^{ab}_{ji}D^{jp}_{qr} \tag{192}$$

$$-2\,g^{bq}_{pr}t^{ab}_{ij}D^{j}_{p}D^{q}_{r} + g^{bq}_{pr}t^{ab}_{ji}D^{j}_{p}D^{q}_{r} \tag{193}$$

$$+g^{bq}_{rp}t^{ab}_{ij}D^{q}_{r}D^{j}_{p} - \tfrac{1}{2}\,g^{bq}_{rp}t^{ab}_{ji}D^{q}_{r}D^{j}_{p}\Big) \tag{194}$$

$$\hat{E}^{a}_{p}\ \Big(-\tfrac{1}{2}\,g^{bp}_{qr}t^{ab}_{ij}D^{i}_{q}D^{j}_{r} - \tfrac{1}{2}\,g^{bp}_{qr}t^{ab}_{ji}D^{ij}_{qr} \tag{195}$$

$$+g^{bp}_{qr}t^{ab}_{ji}D^{i}_{q}D^{j}_{r}\Big) \tag{196}$$

$$\hat{E}^{pa}_{ij}\ \Big(-g^{bq}_{pr}t^{ab}_{ij}D^{q}_{r} + \tfrac{1}{2}\,g^{bq}_{rp}t^{ab}_{ij}D^{q}_{r}\Big) \tag{197}$$

$$\hat{E}^{pa}_{qi}\ \Big(-\tfrac{1}{2}\,g^{bq}_{pr}t^{ab}_{ij}D^{j}_{r} + \tfrac{1}{2}\,g^{bq}_{pr}t^{ab}_{ji}D^{j}_{r} \tag{198}$$

$$+g^{bq}_{rp}t^{ab}_{ij}D^{j}_{r} - \tfrac{1}{2}\,g^{bq}_{rp}t^{ab}_{ji}D^{j}_{r}\Big) \tag{199}$$

$$\hat{E}^{pq}_{ij}\ \Big(\tfrac{1}{2}\,g^{ab}_{pq}t^{ab}_{ij}\Big) \tag{200}$$

$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right] = \hat{E}^{a}_{i}\ \Big(-2\,f^{b}_{j}t^{b}_{i}t^{a}_{j}\Big) \tag{201}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^{\dagger},\,\hat{T}_1\right]$ **(4 terms, 4 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^{\dagger},\,\hat{T}_1\right] = \hat{E}^{a}_{b}\ \Big(f^{i}_{j}t^{b}_{j}t^{a}_{i}\Big) \tag{202}$$

$$\hat{E}^{a}_{p}\ \Big(-f^{b}_{p}t^{b}_{i}t^{a}_{i}\Big) \tag{203}$$

$$\hat{E}^{i}_{j}\ \Big(f^{a}_{b}t^{a}_{i}t^{b}_{j}\Big) \tag{204}$$

$$\hat{E}^{p}_{i}\ \left(-f^{j}_{p}t^{a}_{j}t^{a}_{i}\right) \tag{205}$$

$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] = \hat{E}^{ab}_{ij}\ \Big(f^{c}_{k}t^{a}_{k}t^{bc}_{ij}\Big) \tag{206}$$

$\left[\left[\hat{F},\,\hat{T}_1\right]^{\dagger},\,\hat{T}_2\right]$ **(2 terms, 2 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_1\right]^{\dagger},\,\hat{T}_2\right] = \hat{E}^{ia}_{jk}\ \Big(-f^{b}_{c}t^{c}_{i}t^{ab}_{jk}\Big) \tag{207}$$

$$\hat{E}^{pa}_{ij}\ \left(f^{k}_{p}t^{b}_{k}t^{ab}_{ij}\right) \tag{208}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right]$ **(1 terms, 1 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = \hat{E}^{ab}_{ij}\ \left(f^{c}_{k}t^{bc}_{ij}t^{a}_{k}\right) \tag{209}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_1\right]$ **(6 terms, 4 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_1\right] = \hat{E}^{ia}_{pb}\ \left(f^{c}_{p}t^{bc}_{ij}t^{a}_{j} - f^{c}_{p}t^{bc}_{ji}t^{a}_{j}\right) \tag{210}$$

$$\hat{E}^{ij}_{ka}\ \left(-f^{b}_{c}t^{ab}_{ij}t^{c}_{k}\right) \tag{211}$$

$$\hat{E}^{ij}_{pk}\ \left(f^{a}_{p}t^{ab}_{ij}t^{b}_{k}\right) \tag{212}$$

$$\hat{E}^{pi}_{ja}\ \left(-f^{k}_{p}t^{ab}_{ik}t^{b}_{j} + f^{k}_{p}t^{ab}_{ki}t^{b}_{j}\right) \tag{213}$$

$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right]$ **(0 terms, 0 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 0 \tag{214}$$

$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right]$ **(38 terms, 9 operator groups)**

$$\left[\left[\hat{F},\,\hat{T}_2\right]^{\dagger},\,\hat{T}_2\right] = \hat{E}^{a}_{b}\ \left(2\,f^{c}_{d}t^{bd}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{l} - f^{c}_{d}t^{bd}_{kl}t^{ac}_{ij}D^{ij}_{kl}\right) \tag{215}$$

$$-f^{c}_{d}t^{bd}_{lj}t^{ac}_{ik}D^{k}_{l}D^{i}_{j} - 2\,f^{l}_{p}t^{bc}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{p} \tag{216}$$

$$+f^{l}_{p}t^{bc}_{jl}t^{ac}_{ki}D^{i}_{j}D^{k}_{p} + f^{l}_{p}t^{bc}_{kl}t^{ac}_{ij}D^{ij}_{kp} \tag{217}$$

$$+2\,f^{l}_{p}t^{bc}_{lj}t^{ac}_{ik}D^{i}_{j}D^{k}_{p} - f^{l}_{p}t^{bc}_{lj}t^{ac}_{ki}D^{i}_{j}D^{k}_{p} \tag{218}$$

$$-f^{l}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{ij}_{kp}\Big) \tag{219}$$

$$\hat{E}^{a}_{i}\ \left(-f^{b}_{p}t^{bc}_{jk}t^{ac}_{il}D^{jk}_{lp} + \tfrac{1}{2}\,f^{b}_{p}t^{bc}_{jk}t^{ac}_{li}D^{jk}_{lp}\right. \tag{220}$$

$$+2\,f^{b}_{p}t^{bc}_{kl}t^{ac}_{ij}D^{j}_{k}D^{l}_{p} - f^{b}_{p}t^{bc}_{kl}t^{ac}_{ji}D^{j}_{k}D^{l}_{p} \tag{221}$$

$$\left.-f^{b}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{l}_{p}D^{j}_{k} + \tfrac{1}{2}\,f^{b}_{p}t^{bc}_{lk}t^{ac}_{ji}D^{l}_{p}D^{j}_{k}\right) \tag{222}$$

$$\hat{E}^{a}_{p}\ \left(\tfrac{1}{2}\,f^{b}_{p}t^{bc}_{jl}t^{ac}_{ik}D^{i}_{j}D^{k}_{l}\right. \tag{223}$$

$$\left.-f^{b}_{p}t^{bc}_{lj}t^{ac}_{ik}D^{k}_{l}D^{i}_{j} + \tfrac{1}{2}\,f^{b}_{p}t^{bc}_{lk}t^{ac}_{ij}D^{ij}_{kl}\right) \tag{224}$$

$$\hat{E}^{ia}_{jb}\ \left(-f^{c}_{d}t^{bd}_{il}t^{ac}_{jk}D^{k}_{l} + \tfrac{1}{2}\,f^{c}_{d}t^{bd}_{il}t^{ac}_{kj}D^{k}_{l}\right. \tag{225}$$

$$\left.+\tfrac{1}{2}\,f^{c}_{d}t^{bd}_{li}t^{ac}_{jk}D^{k}_{l} - f^{c}_{d}t^{bd}_{li}t^{ac}_{kj}D^{k}_{l}\right. \tag{226}$$

$$+f_p^l t_{il}^{bc} t_{jk}^{ac} D_p^k - \tfrac{1}{2}\, f_p^l t_{il}^{bc} t_{kj}^{ac} D_p^k \tag{227}$$

$$-f_p^l t_{li}^{bc} t_{jk}^{ac} D_p^k + \tfrac{1}{2}\, f_p^l t_{li}^{bc} t_{kj}^{ac} D_p^k \Big) \tag{228}$$

$$\hat{E}_{jk}^{ia} \Big(f_p^b t_{il}^{bc} t_{jk}^{ac} D_p^l - \tfrac{1}{2}\, f_p^b t_{li}^{bc} t_{jk}^{ac} D_p^l \Big) \tag{229}$$

$$\hat{E}_{pj}^{ia} \Big(\tfrac{1}{2}\, f_p^b t_{il}^{bc} t_{jk}^{ac} D_l^k \tag{230}$$

$$-\tfrac{1}{2}\, f_p^b t_{il}^{bc} t_{kj}^{ac} D_l^k - f_p^b t_{li}^{bc} t_{jk}^{ac} D_l^k \tag{231}$$

$$+\tfrac{1}{2}\, f_p^b t_{li}^{bc} t_{kj}^{ac} D_l^k \Big) \tag{232}$$

$$\hat{E}_{kl}^{ij} \Big(f_b^a t_{ij}^{ac} t_{kl}^{bc} \Big) \tag{233}$$

$$\hat{E}_{ib}^{pa} \Big(-\tfrac{1}{2}\, f_p^l t_{kl}^{bc} t_{ij}^{ac} D_k^j + f_p^l t_{kl}^{bc} t_{ji}^{ac} D_k^j \tag{234}$$

$$+\tfrac{1}{2}\, f_p^l t_{lk}^{bc} t_{ij}^{ac} D_k^j - f_p^l t_{lk}^{bc} t_{ji}^{ac} D_k^j \Big) \tag{235}$$

$$\hat{E}_{jk}^{pi} \Big(f_p^l t_{il}^{ab} t_{jk}^{ab} \Big) \tag{236}$$